

Technical potential appraisal and optimal site screening comparing AHP and fuzzy AHP methods for large-scale CSP plants: A GIS-MCDM approach in Morocco

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ABSTRACT

Given that Direct Normal Irradiation (DNI) levels are consistently elevated, Concentrating Solar Power (CSP) technology is a highly viable alternative in Morocco enabling reliable power generation. Herein, an assessment of the solar CSP technical potential was initiated, along with a comparative site suitability analysis for CSP plants using two methods: Analytic Hierarchy Process (AHP) and Fuzzy AHP. Findings brought to light Morocco's sizeable potential reaching 21,100.32 TWh/year. Turning to distribution, most regions are appropriate, notably southern ones. Laâyoune-Sakia El Hamra and Dakhla-Oued Ed-Dahab regions rank first as regards potential, reaching 5,549.72 and 5,117.31 TWh/year accordingly. Thereafter, site screening to draw comparisons between AHP and FAHP methods was established depicting discrepancies in weighting and ranking maps. Consistent with literature, screening criteria were structured by climate, orography, proximity to water resources and location. Weightings were calculated, revealing roughly the same results, with minor variations. Climate was ranked first (57.72 % for FAHP with α -cut, 57.34 % for FAHP with CoA, 57.70% for AHP) and location was last (5.04% for FAHP with α -cut, 5.07 % for FAHP with CoA, 5.00% for AHP). Ranking maps and area ratio calculations allowed variation highlight. The α -cut defuzzification provides closer approximation to subjectivity than the CoA method.

Introduction

Energy demand rates, currently and in the foreseeable future, are pointing to a handful of critical issues needing to be tackled to effectively accommodate the growing population's quest for a higher standard of living [1]. Alongside that, mankind's activities have driven global warming, mainly by releasing greenhouse gases, raising the global surface temperature to 1.1 °C over the 1850–1900 period in 2011–2020. Deep and rapid transitions are crucial in all sectors and across all systems to guarantee a viable and sustainable future for all people [2].

Relying solely on fossil fuels is no longer the optimal nor the best option for energy production. The harnessing of renewable energies is an established trend, as several countries have embarked on the planning, implementation and operation of solar plants and wind farms, alongside other renewables such as geothermal energy. For instance, countries such as China, India and the European Union have pledged to achieve

promising renewable energy (RE) targets [3]. Morocco in turn has initiated, since 2009, a national energy strategy seeking to increase the share of renewable energies (wind, solar, hydro) in the energy mix to 52 % by 2030 in terms of total installed capacity [4,5].

Within the Moroccan context, a comprehensive screening of all renewable alternatives, with a view to fine tuning future installations, might be not only a promising path towards the 52 % target outlined above, but also toward more sensible deployment of solar power plants. Solar energy is a key component of the mix represented by photovoltaic (PV) and concentrating solar power (CSP) technologies. As for PV technology, Ouchani et al [6] explored both the technical and economic potentials with regard to the large-scale deployment of solar PV plants in Morocco. The implementation limits have been taken into account by ruling out inappropriate areas. Roughly 87.52 % by area was deemed suited to PV installations. In the most optimistic scenario, the total technical potential amounted to 49,643 TWh/year.

At the same time, Concentrating solar power (CSP) holds great

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promise as a low-emission power generation technology, since it uses the heat of the sun to prevent the combustion necessary for the majority of mainstream power systems [7]. Owing to the high values of Direct Normal Irradiation (DNI) [8], the CSP technical potential is worth assessing in Morocco. Accordingly, the first section of this work will focus on the evaluation of the technical potential.

Generally, potential evaluation is conducted in order to assess the energy that could be produced with regard to the systems constraints and conversion ratios nowhere near the theoretical potential. In this regard, several studies have focused on this topic, in particular Chen et al [9] in China, Fluri [10] in South Africa, and Purohit et al [11] in India. Essentially, the potential assessment provides a rough estimate of the amount of energy which might be produced and its value before proceeding to the site selection analysis.

Given that renewable energy technologies tend to consume a lot of land locally, a holistic energy-planning approach may contribute to assuring that RE technologies are deployed efficiently with minimal detrimental effects [12].

Literature review on comparative AHP and FAHP analysis

As such, multiple solar siting studies have focused on selecting appropriate sites with the aim of evaluating and classifying them to find optimal locations (either PV or CSP). Although PV technology seems to have been widely researched [13–20], several studies have sought to address CSP technology [8,21–25] too. In the meantime, other investigations have considered both technologies [26–28].

Deciding upon the location of a solar plant is a critical planning decision [29], the site itself needs to fulfill many conditions related to the constraints of the technology and other restrictions that may hinder the actual installation. The aforementioned potential studies took account of a number of criteria impacting the location using Multi Criteria Decision Making (MCDM) methods, with a view to assigning weights to the criteria, along with Geographic Information System (GIS) allowing the generation of the final ranking maps displaying the distribution of chosen locations.

Various Multi Attribute Decision Making (MADM) methods are available and widely considered in the literature, where each of them has its own characteristics. Among these, nevertheless, some methods have a higher potential to solve decision making related problems, namely the following: Weighted Product Method (WPM), Weighted Sum Method (WSM), Analytic Hierarchy Process (AHP), Analytic Network Process (ANP), Elimination and Choice Translating Reality (ELECTRE), Preference Ranking Organization Method for Enrichment of Evaluations (PROMETHEE), Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), ViseKriterijumska Optimizacija i Kompromisno Resenje (VIKOR), and Decision Making Trial and Evaluation Laboratory (DEMATEL) [30]. Meanwhile, as far as the assessment of RE investments and/or projects is concerned, the AHP, ANP, ELECTRE and TOPSIS methods appear to be the most commonly used in the literature [31]. Among these selected methods, AHP stands out as the most widely applicable for various types of problems and remains the most successful and popular among the MCDM methods. Besides, it has been broadly adopted for the assessment as well as the selection of technologies in the RE segment [31,32].

The Analytic Hierarchy Process (AHP) was one of the most used MCDM methods in such studies [15,16,25,33,34]. The latter method is always described as subjective, since the pairwise comparison is conducted by individuals. Given that this is its only flaw, many research papers have countered this problem by consulting a group of experts prior to aggregating their judgments [35]. Meanwhile, other researchers have opted for the FAHP since the fuzziness reduces the subjectivity and thus ensures more accuracy [36].

As a matter of fact, various studies have compared the two previously mentioned approaches, albeit for different applications aside from site selection studies. Tripathi et al [37] have conducted a study which

purpose was to define the most suitable MCDA method for the selection of new hospital sites. To this end, eleven criteria were selected, classified according to three major aspects: socioeconomic, geographic, and environmental. Two MCDA methods, namely the Analytic Hierarchy Process (AHP) and the Fuzzy Analytic Hierarchy Process (FAHP), were applied in combination with GIS tools. Through comparison, the difference of these methods in assigning the weight of the criteria is ultimately reflected in the ranking of the respective criteria. Findings imply that the FAHP method may provide a more effective way to approach the hospital siting problem.

As for Mosadeghi et al [38], the use of quantitative approaches such as multi-criteria decision-making techniques in land suitability cases has been expanding, enabling heterogeneous data to be managed. Nevertheless, the study demonstrates that spatial decision making can be utilized to prioritize options and perform scenario analyses, as well as to provide insight into the spatial range of alternatives. Toward this end, the paper compared the results of two techniques (AHP) and Fuzzy AHP in the determination of land use area size in a large-scale urban planning scheme. The outcome illustrates that in the initial stages of the planning procedure, when development options need to be identified as a focal point, rather simplified methods can be enough. But, in cases where planning necessitates defining the spatial size of the favored development areas, the intersection area suggested by the two methods will be optimal to use.

On another note, Khashei-Siuki et al [39] have compared the two previously mentioned MCDM methods to determine appropriate areas for drinking water harvesting. For this matter, ten parameters were used as decision criteria. Their values were calculated using the AHP and FAHP methods. Depending on the effect of each parameter value on the raster layer, all GIS raster layers were combined into two final ones. The final FAHP layer was more consistent with reality than the final AHP layer.

After highlighting these various applications, the intention of this paper is to assess the technical potential of solar thermal in Morocco as a supporting decision aid for solar installations especially that PV technical potential has already been previously assessed by Ouchani et al. [6]. Afterwards, a comparative analysis of AHP and fuzzy AHP methods is conducted in an energy perspective with regard to varying defuzzification methods and their impact, all combined with GIS for CSP site selection. Additionally, this paper seeks to illustrate the extensive surface area suited to CSP plants, while also highlighting the tremendous potential and adaptability of several areas for solar thermal power plants. A display of the fuzziness effect on the assessment of suitable sites for solar thermal power plants is investigated to compare ratio area at last.

The novelty of this work resides in consolidating the renewable potential framework by appraising the potential of CSP in Morocco, with a careful consideration of the variation of solar thermal efficiency for a high level of accuracy and a more realistic evaluation. Besides, the comparison between AHP and FAHP in an energy context is sought and can give other insights with the visualization of the variations is beneficial for future studies that would allow researchers a proper allocation of decision making methods.

The present research paper leads off with the first section covering the approach and methods, to reach the zone of interest, leading on to the process of data analysis and processing for comparison purposes. The subsequent section scrolls through outcomes, first assessing the solar technical potential, followed by prospecting for appropriate locations for CSP, with ranking maps derived by means of both the AHP and Fuzzy AHP methods (with CoA and α -cut defuzzification methods), together with a comparative analysis of the respective weight tables and final maps. The work concludes with a general overview of some noteworthy key points.

Methods and approach

CSP technical potential approach

Within the assessment of the potential of renewable energy sources, a distinction is made between a number of various classes requiring a sequence of handling steps and hypotheses to be made. Transitioning from theoretical to technical potential involves the inclusion of a series of further data assumptions, including geographic potential as an intermediate phase [40].

The theoretical potential refers to the quantity of resource that exists regardless of the efficiency of conversion and losses; it corresponds to the highest available physical energy from a given source. With regard to solar energy in particular, the latter potential refers to the total solar radiation reaching the surface being evaluated [41]. The geographic potential is taken as a stepping stone to calculate the technical potential at last. It covers zones suited to and capable of being used for certain applications in the field of renewable energy. According to the accessible data, the restricted areas are considered and ruled out in order to assess the remaining exploitable areas (e.g. urban areas, steep areas, water bodies, protected areas and forests and so on) [40]. Subsequently, the technical potential represents the geographic potential subtracting conversion losses related to secondary energy and limited by installation constraints on a large scale (for instance spacing factors reflecting the spacing and service zones of solar power plants along with the transmission lines losses) [40,42]. Further levels such as economic and market potentials can be explored to enhance the assessment from other perspectives.

Unlike PV installations, the connection between resource (irradiation) and output is somewhat different in the case of CSP technology due to the higher efficiency of CSP systems at increased irradiation levels. The technical potential (TPot) can therefore be represented as follows according to Hermann et al. [40]:

$$TPot_{total} = \sum_{i=1}^n TPot_i = \sum_{i=1}^n \left(RA_i \times \frac{Eff_i}{SF} \times A_i \right) \quad (1)$$

where $TPot_{total}$ stands for the total technical potential value (GWh/year), $TPot_i$ is the technical potential in cell i (GWh/year), RA_i is the resource availability ($kWh/m^2/year$), Eff_i represents the CSP plant efficiency (%), SF is the spacing factor, A_i is the available area (km^2) and n the number of pixels.

The overall technical potential amount is considered as the sum of TPot in each cell i . RA_i is the resource availability referring to the DNI value considered the best estimate of final output, with CSP as the target

technology. The Eff_i factor – CSP plant efficiency – is plotted as a function of irradiation as indicated in Fig. 1. In compliance with Hermann et al. [40], various efficiency values were taken into account with regard to the amount of irradiation in a given area. Accordingly, efficiency ranges were set from around 12 % for irradiation values of 1800 $kWh/m^2/year$ rising up to a peak of 16 % for 4000 $kWh/m^2/year$. Throughout calculations, the efficiencies associated with each cell were derived from the curve illustrated in Fig. 1, on the basis of the corresponding DNI value.

The Spacing Factor (SF) serves to assess the effective use of land compared to the area of the panels (for PV) or mirrors (for CSP). An average parameter for CSP installations amounts to between 5 and 10 (e.g., the required ground area is 5 to 10 times greater than the effective area “capturing” irradiation). On the basis of Hermann et al [40], the retained value was 7.5. As for A_i , it represents the available surface area estimated on the basis of GIS assessment, with account taken of excluded zones.

For the sake of depicting zones to be discarded, a series of selective restrictions need to be processed according to existing literature. If some of the latter restrictions might be similar for all the existing RE systems regarded for evaluation, others differ due to technical requirements regarding the field and the installed systems.

As for the common ones, residential areas are omitted since the solar thermal plants are large-scale, as well as forests and protected areas along with water bodies. Also, military zones are considered as sensitive areas to be ruled out. When it comes to the limitations related to the solar thermal technology, under-resourced areas –with DNI values beneath 1800 $kWh/m^2/year$ – and with slopes exceeding 2.1° do not allow a good performance of the technology and might rise the investment cost.

Fuzzy set theory in MCDM and fuzzy AHP method

For many situations, precise data is unsuited as a way of describing life situations given that evaluations and human preferences exhibit a high degree of subjectivity, uncertainty and ambiguity, and therefore are not measurable by means of exact numerical values [43]. The theory of fuzzy sets originated with Zadeh (1965, 1976) [44,45] to address the inaccuracy and ambiguity within both individual cognitive and human decision making, it allows data to be manipulated through mathematical computational forces to handle the uncertainties typically encountered in the thinking and reasoning of humans, and by allowing membership of partial sets rather than precise sets. Some implementations of fuzzy theories to various decision-making operations in an uncertain

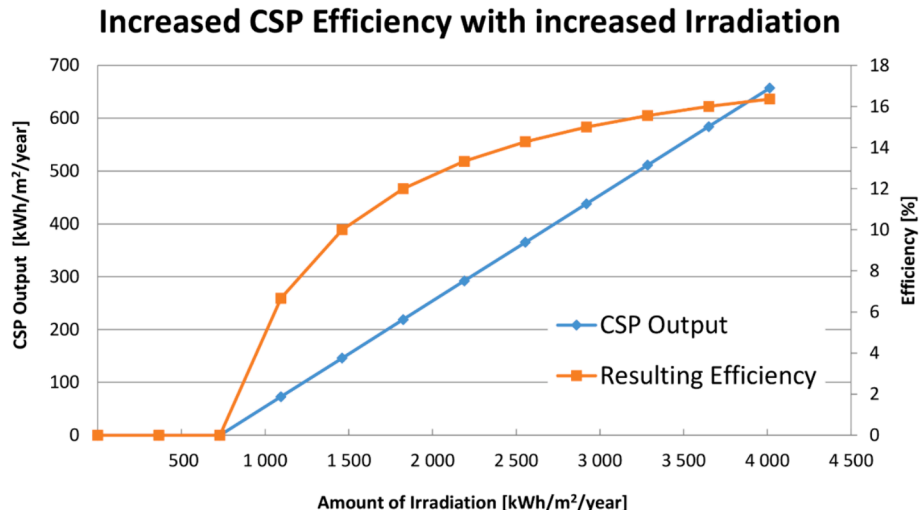


Fig. 1. CSP efficiency as a function of irradiation receiving according to Herman et al. [40].

environment were presented by Bellman and Zadeh (1970) [46], who introduced the fuzzy MCDM methodology [43].

Fuzzy Analytic Hierarchy Process (FAHP) approach using geometric mean for weight calculation (Buckley 1985) is one among many other FAHP methods developed by numerous authors, such as Van Laarhoven and Pedrycz (1983), Chang (1996), Mikhailov (2000) and (2003) and others. Whilst it results in fuzzy weights requiring defuzzification, the chosen method brings a number of advantages, as it is easy to implement, and to extend to the fuzzy case, as well as ensuring a unique solution [36,47].

Buckley (1985) has initially conceived the geometric mean method in order to enlarge the AHP to the situation of the use of linguistic variables [48]. The pairwise comparison of these variables in terms of degree may be described through fuzzy numbers, as Table A1 outlines [36].

As defined by Laarhoven and Pedrycz (1983), a triangular fuzzy number (TFN) is required to have the main following characteristics [48]. A fuzzy number $\tilde{A}$ on $\mathbb{R}$ is a TFN (see Fig. 2) if its membership function $x \in \tilde{A}$, $\mu_{\tilde{A}}(x) : \mathbb{R} \rightarrow [0, 1]$ is equal to

$$\mu_{\tilde{A}}(x) = \begin{cases} \frac{(x-l)}{(m-l)}, & l \leq x \leq m \\ \frac{(u-x)}{(u-m)}, & m \leq x \leq u \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where l and u respectively represent the lower and upper bounds of the fuzzy number $\tilde{A}$, and m the modal value [48].

The first step consists of establishing the fuzzy pairwise comparison matrix $\tilde{A}$ as following, using TFNs mentioned in Table A1:

$$\tilde{A} = \begin{bmatrix} (1, 1, 1) & \tilde{a}_{12} & \cdots & \tilde{a}_{1n} \\ \tilde{a}_{21} & (1, 1, 1) & \cdots & \tilde{a}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{a}_{n1} & \tilde{a}_{n2} & \cdots & (1, 1, 1) \end{bmatrix}$$

where $\tilde{a}_{ij} \times \tilde{a}_{ji} \approx 1$ and $\tilde{a}_{ij} \cong w_i/w_j$ with $i, j = 1, 2, \dots, n$.

The second step is the calculation of the fuzzy geometric mean value $\tilde{r}_i$, for each criterion i which is computed as

$$\tilde{r}_i = (\tilde{a}_{i1} \times \tilde{a}_{i2} \times \cdots \times \tilde{a}_{in})^{1/n}$$

After that, the fuzzy weight $\tilde{w}_i$ for each criterion i is calculated as

$$\tilde{w}_i = \tilde{r}_i \times (\tilde{r}_1 + \tilde{r}_2 + \cdots + \tilde{r}_n)^{-1}$$

where $\tilde{r}_k = (l_k, m_k, u_k)$ and $(\tilde{r}_k)^{-1} = (1/u_k, 1/m_k, 1/l_k)$

Then, the fuzzy weights $\tilde{w}_i = (l_i, m_i, u_i)$ are defuzzified with the aid of any defuzzification method. There are two major defuzzification methods used by researchers for type-1 fuzzy sets (TFN and TraFN): center of area method (CoA) and extent analysis method [51]. Also, the

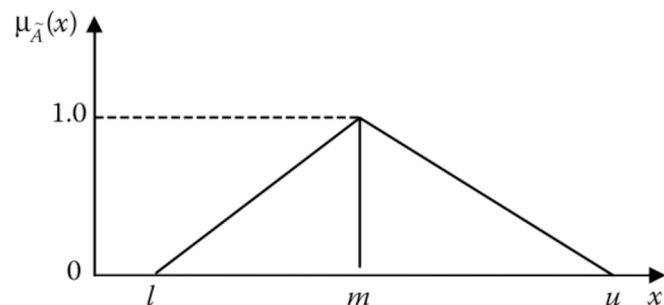


Fig. 2. Membership function of the TFN [48].

α -cut method has been amongst the remaining used defuzzification methods [51,52]. In our case, the center of area (CoA) method was used as mentioned below [49]:

$$w_i = \frac{l_i + m_i + u_i}{3}$$

Meanwhile the α -cut method allows us to calculate the crisp weight for TFNs as following [50]:

$$w_i = \frac{(l_i + 4 \cdot m_i + u_i)}{6}$$

Regarding consistency checking, calculation of the consistency ratio (CR) is based on Saaty's approach, but with careful consideration of fuzzy pairwise comparisons as outlined by Gogus and Boucher [53]. When using TFNs, consistency ratios for the matrix of mean values (CR^m) and the matrix of geometric means of the lower and upper limits (CR^g) are required.

To this end, A^m and A^g from a single matrix of fuzzy pairwise comparisons of $n \times n$ ($\tilde{A}$) are set. For the first matrix, the mean values of the preferences are simply entered, $A^m = [a_{ijm}]$. While the second matrix is made up of the geometric means of the lower and upper limits, $A^g = [\sqrt{a_{iju} \cdot a_{ijl}}]$.

Finding the consistency rate involves calculating the weight vectors for each matrix. As each matrix is made up of crisp data, Saaty's technique for finding the weight vectors works well. Weight vectors w^m and w^g using the following formula:

$$w^m = [w_i^m] \text{ where } w_i^m = \frac{1}{n} \sum_{j=1}^n \frac{a_{ijm}}{\sum_{i=1}^n a_{ijm}}$$

$$\text{and } w^g = [w_i^g] \text{ where } w_i^g = \frac{1}{n} \sum_{j=1}^n \frac{\sqrt{a_{iju} \cdot a_{ijl}}}{\sum_{i=1}^n \sqrt{a_{iju} \cdot a_{ijl}}}$$

Thereafter, the largest eigenvalue of each matrix is expressed as follows:

$$\lambda_{max}^m = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n a_{ijm} \left(w_j^m / w_i^m \right),$$

$$\lambda_{max}^g = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n \sqrt{a_{iju} \cdot a_{ijl}} \left(w_j^g / w_i^g \right)$$

According to Saaty's approach, the consistency indices (CI) and ratios are derived by means of the formulae below and RI values in Table 1

$$CI^m = \frac{(\lambda_{max}^m - n)}{(n - 1)}, \quad CI^g = \frac{(\lambda_{max}^g - n)}{(n - 1)}$$

$$\text{and } CR^m = \frac{CI^m}{RI^m}, \quad CR^g = \frac{CI^g}{RI^g}$$

As regards TFNs, both CR^m and CR^g shall be determined, whereby a value of 0.1 or less is viewed as valid and satisfactory [53,54].

Table 1
Random indices [53].

Size of the matrix	RI ^m	RI ^g
1	0	0
2	0	0
3	0.4890	0.1796
4	0.7937	0.2627
5	1.0720	0.3597
6	1.1996	0.3818
7	1.2874	0.4090
8	1.3410	0.4164
9	1.3793	0.4348
10	1.4095	0.4455
11	1.4181	0.4536
12	1.4462	0.4776
13	1.4555	0.4691
14	1.4913	0.4804
15	1.4986	0.4880

Area of interest

At the present time, the Kingdom of Morocco relies heavily on imports for over 91.31 % of its energy needs [55]. In line with the energy strategy, the current assessment viewed Morocco as an area of interest (Fig. 3) in the light of its substantial solar potential, particularly solar technologies, which could be exploited to generate electricity from renewable energies, thereby helping reduce energy dependency. Indeed, the hours of sunshine fall within the range of 2700 to more than 3500 per year with an irradiation of about 5.3 kWh/m²/day [56]. As a matter of fact, DNI values appear to be reaching a maximum of 3232 kWh/m²/year.

Geographically, Morocco is located within the ranges of latitude 20.76 to 35.92 decimal degrees (DD) and longitude −0.99 to −17.10 (DD), in the northwest of Africa. The kingdom shares a border to the south with Mauritania, and to the east with Algeria. In addition, it is surrounded to the north by the Mediterranean Sea and to the West by the Atlantic Ocean. With a total area of 710,850 km² [57], the country is projected to reach a population of 66.4 million by 2030 [58]. Depending on its topography, the climate in Morocco varies with the Rif Mountains on the northern side, the Atlas Mountains in the central part, the plateaus on the east, the plains and coast on the west, and the desert on the southern side. Nearly all of Morocco, especially near its shores, has a typical Mediterranean climate – with mild and wet winters as well as hot and dry summers-. The Atlas Mountains traverse central Morocco, marking a natural split separating the northern Mediterranean coastal zone from the southern inland regions bordering the Sahara Desert [58].

MCDM-GIS approach: From restriction to weighting

Restriction analysis

The present analysis is carried out with the aim of pinpointing and narrowing down locations incapable of accommodating solar thermal power plants, with due regard to a range of constraints. Indeed, amidst site selection studies for solar plants – CSP technology –, it is compulsory to build up a specific mask covering the unsuitable areas. The latter ones extend beyond the abovementioned zones (restricted areas for technical potential) with additional constraints. A 500 m buffer zone is included around waterbodies (rivers, dams, lakes). Additionally, the infrastructure is regarded mainly as roads and railways with a buffer of 100 m. Also, airports with a 2500 m buffer are removed as they have a negative effect on aircraft flight caused by sunlight reflection [59]. Moreover, military zones and protected areas as well as forests with a 500 m buffer are discarded.

The aforementioned layers are either retrieved from [60] or digitalized, and buffers were generated thereafter. The merge was then executed to form the mask containing all the restricted zones. As a final display, the map –Fig. 4– portrays the layers gathered for our exclusion.

Decision-making criteria

With regard to the screening criteria, a variety of factors were taken into account in a multi-criteria decision-making process which enabled ranking of the criteria with respect to their relevance. The criteria retained for the present case study are primarily representative of the valid literature. Four main groups have been identified: Climate, orography, water resources and location.

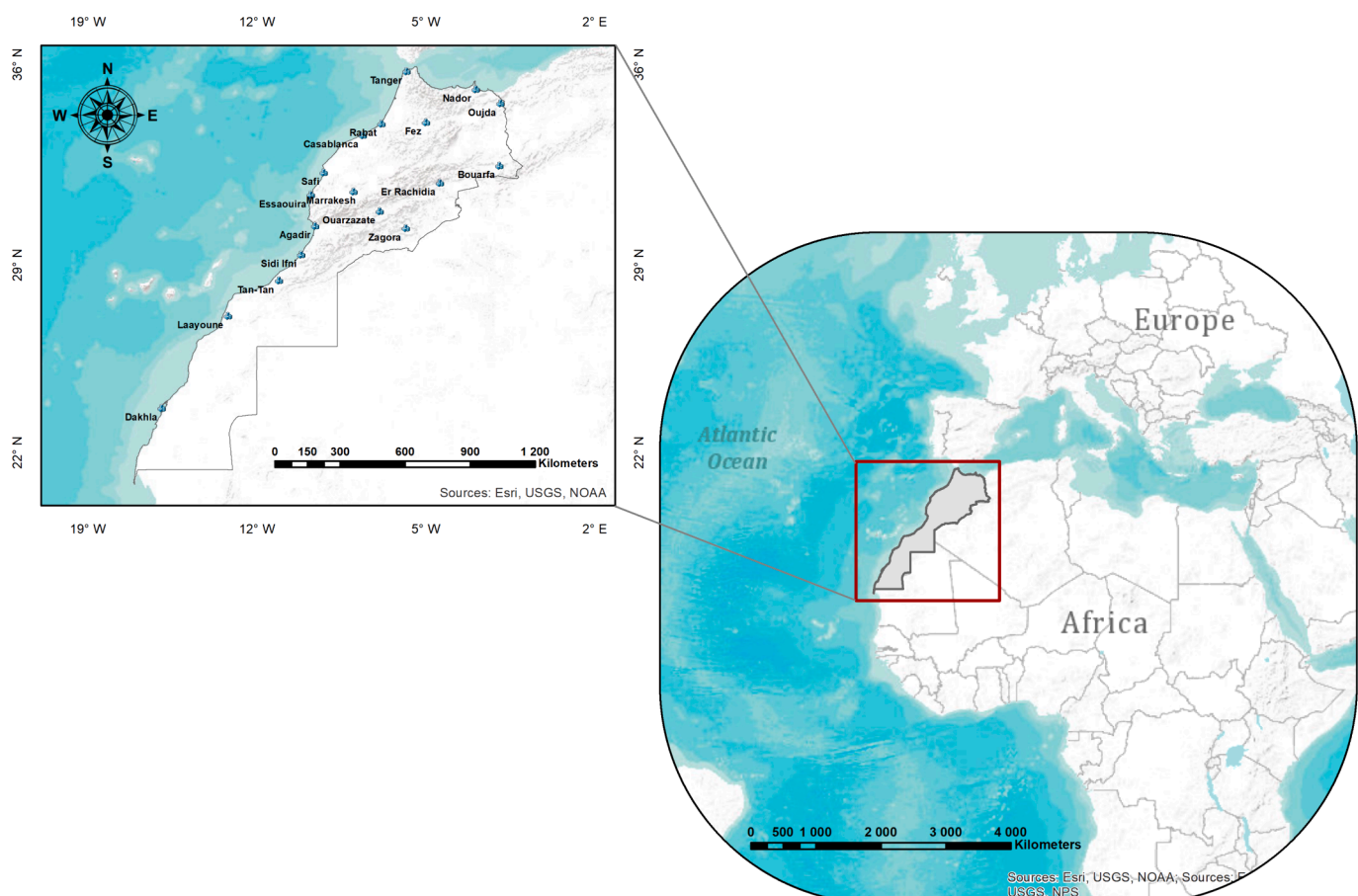


Fig. 3. Area of interest location.

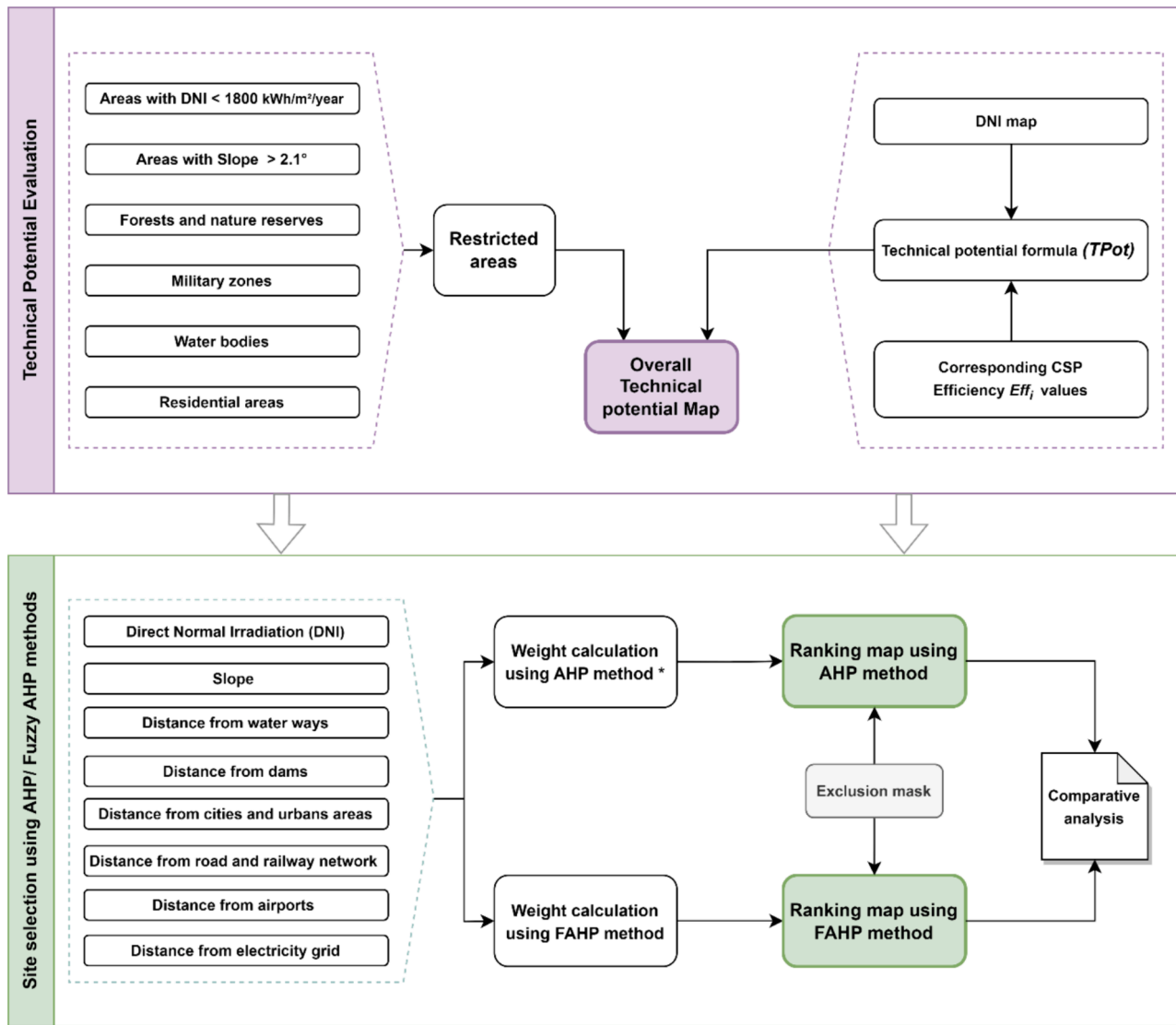


Fig. 4. General outline of this research * (Weighting results retrieved from Tazi et al [61]).

In the “Climate” category, the main criterion is the DNI, which is essential and directly affects the system’s performance and hence the power plant’s output. As such, it is key to determining whether an investment in CSP technology is worthy of consideration, as DNI can provide insight into not only the size of the thermal power plant, but also its performance and the land area it can occupy [62]. A satellite-derived solar resource data set was analyzed over a twenty-year period (1991 to 2010) [63]. Aside from taking slope into account, the latter map is a high-resolution raster data where each pixel is 1 km². The present DNI map was provided by “the Solar Atlas for the Mediterranean” project conceived in the context of project launched by MINES ParisTech, the German Aerospace center (DLR), ARMINES, as well as TRANSVALOR entitled Solar-Med-Atlas [64]. The minimum required value to ensure satisfactory output performance has been set at 1800 kWh/m²/year from a techno-economic viewpoint, as adopted by Hermann et al [40], Aly et al [27] and also Alami Merrouni et al [25].

As far as orography is concerned, slope was chosen as solely representing the second category with due regard to sensitivity of CSP technology to ground slope. Also, unlike photovoltaic systems, slope orientation has little influence on site selection [25]. There has been no consensus on a precise threshold for the slope, as mentioned by Yushchenko et al [26] many values were taken into account ranging from 1 % to 5 % depending on the chosen technology and Ziuku et al [21] have

tolerated a slope value of up to 7 %. Within this case, the areas greater than 2.1° have been excluded, as considered by [8,40] to prevent any earthwork for the power plant. The Shuttle Radar Topography Mission (SRTM) data was retrieved from [64] and processed using ArcGIS tools to deduce the slope map in Fig. 5 (b).

Regarding water resources, this category is also of a great importance assuming it is only wet-cooled CSP thermal power plants under consideration. Both the cooling system and cleaning and maintenance processes are water intensive [65]. In this particular case, the water ways and the dams were the ones taken into account. Distance from water holds an important bearing on the ranking of sites: the nearer the sites are to the water the better. The processed layers were extracted from [66].

With respect to the last category – Location – in view of large-scale solar thermal plants, siting is oriented towards securing optimal positioning where costs can be reduced during both the construction and operating phases. Proximity to residential areas was regarded first, as it guarantees the access to replenishment and vicinity of qualified workforce. The second one was proximity to roads and railways for easier access and lower costs. Another sub-criterion was proximity to airports as a means of protecting flight safety [67]. The above-mentioned data layers were successfully excerpted from [66] and processed using GIS tools. Also, the closeness to electricity lines is a factor not to be neglected

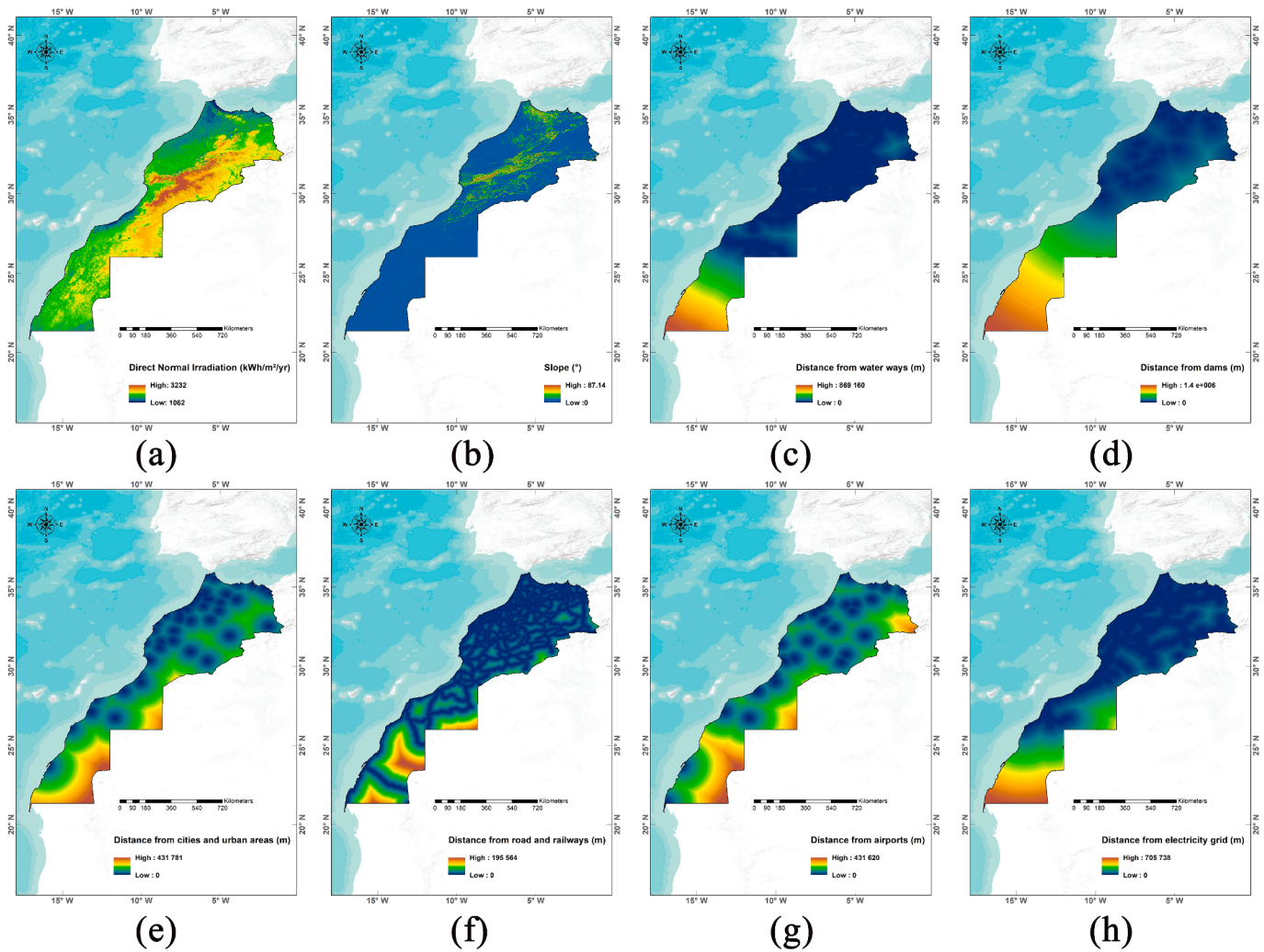


Fig. 5. Criteria layers (a) Direct normal irradiation map, (b) Slope, (c) & (d) Waterways and dams, (e) Cities, (f) Roads and railways, (g) Airports, (h) Electricity grid.

to ease the connection to the existing network after plants installation [21]. The electrical grid data was originally from [64]. All in all, Fig. 5 encompasses each of the criteria layers having been depicted and handled.

Results & Discussion

Based on a careful examination of the availability of suitable land, the first objective of this study was to provide an in-depth estimate of the technical potential of CSP solar, emphasizing the vast potential for harnessing resources among existing renewable alternatives. Moreover, the effect of fuzziness in terms of spatial distribution was examined using the Analytic Hierarchy Process (AHP) and fuzzy AHP as part of site screening. Beyond defining the appropriate candidate sites available, the ultimate goal was to compare the above-mentioned methods in order to show that a slight difference in weighting can have an impact on the final display of the ranking maps. The following sections will address results regarding potential assessment as well as site selection with comparative analysis from two perspectives: weight calculation and ranking maps (spatial distribution).

Assessment of CSP technical potential

On basis of Eq. (1), the estimation was carried out on each cell (i) (a resolution of 1 km^2) with the assistance of ArcGIS software. Indeed, after extracting the corresponding DNI value of each cell, the corresponding efficiency value Eff_i value has been derived. In parallel, an analysis of the remaining available land to be processed has been compiled. The restricted areas were displayed in Fig. 6 depicting the various discarded surfaces.

After processing, the resulting map of technical potential was compiled as illustrated in Fig. 7 which indicates the amount of potential per cell. The latter vary from a minimum of 0.20 to a maximum of 47.97 GWh/year.

The consolidated results demonstrate that Morocco holds significant CSP technical potential up to 21,100.32 TWh/year. This latter amount is on a par with the published IRENA assessment of 21,553 TWh/year [40]. The difference resides within a higher resolution of data utilized within this study, where the derived efficiency varies depending on the DNI. In parallel, it is noticeable that CSP technical potential is lower than the PV technical potential assessed in Ouchani et al. [6] due to the fact that the CSP technology is rather land-intensive.

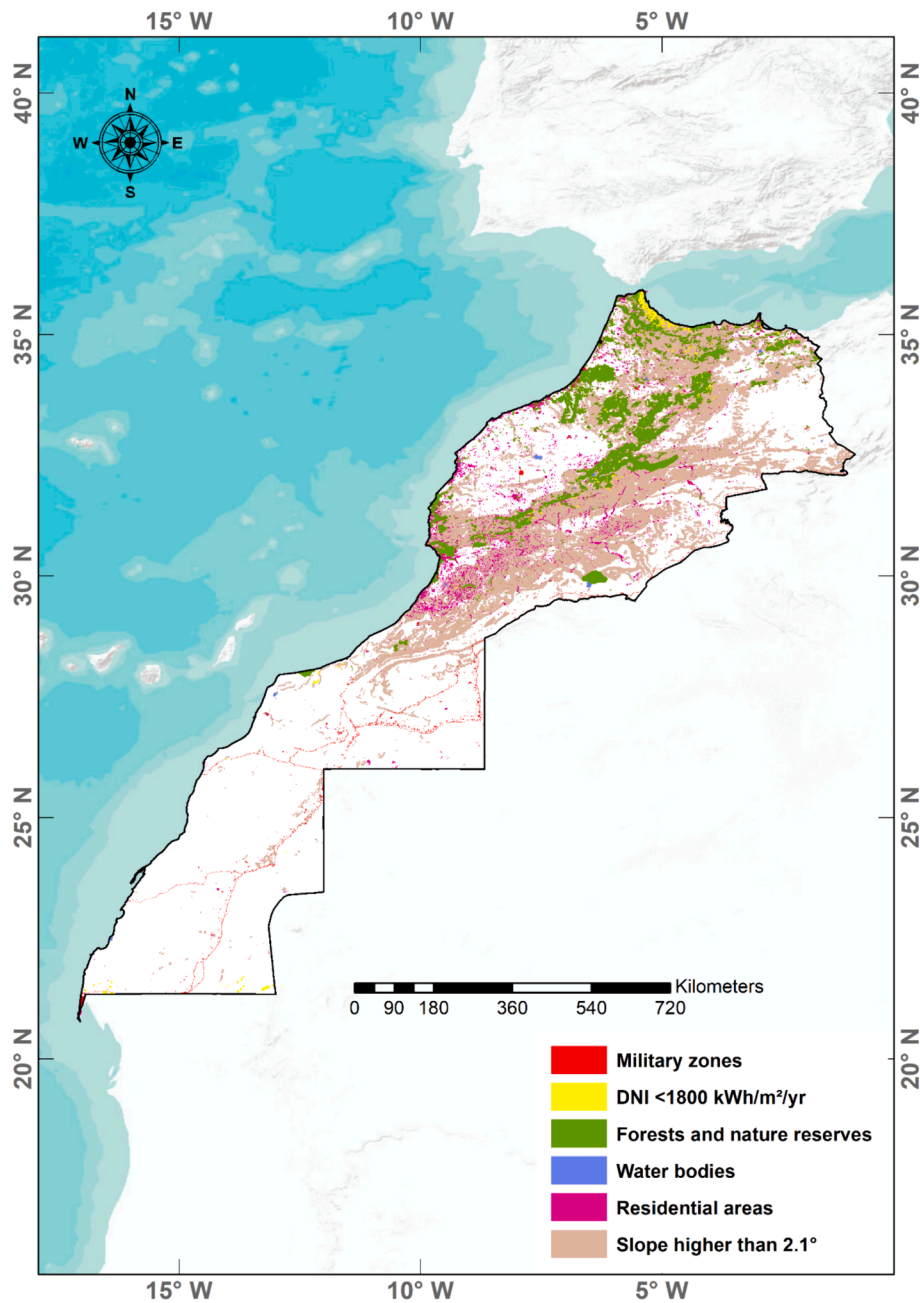


Fig. 6. Layers of the excluded areas

At a regional level, the majority of the areas are highly suitable namely the southern ones, few in the center and also in the east. Whereas the northern zones were less favorable for these kinds of installations mainly due to lower DNI values.

The region of Laâyoune-Sakia El Hamra and Dakhla-Oued Ed-Dahab hold dominant shares of the country's potential rising to 5549.72 and 5117.31 TWh/year respectively. Meanwhile, regions with the lowest potential were Rabat-Salé-Kénitra and Tanger-Tétouan-Al Hoceïma reaching only 375.60 and 127.15 TWh/year correspondingly.

Within the country, five regions gather a little less than three quarters of the total households: Casablanca-Settat, Rabat-Salé-Kénitra, Marrakesh-Safi, Fès-Meknès and Tanger-Tétouan-Al Hoceïma [68]. All the above-mentioned regions have insufficient potential to meet their needs if solar thermal power plants are installed, while the remaining regions have a relatively higher potential, but they have fewer

inhabitants and more available space. Indeed, Table 2 summarizes the calculated potential in each region arranged according to their potential from highest to lowest value. Also, maximum and average DNI are displayed to give a glimpse about irradiation variation and distribution.

With the same logic, there was also consideration of the areas that could be exploited in terms of distribution as well as the share of the total technical potential (see Table 3). The estimations reflect the availability of land to be possibly exploited and the surface that could be narrowed with more refinement. The southern regions have been found to be the most advantageous and hold approximately 58.11 % of the total technical potential. Conversely, the regions offering lower percentages and hence proving less interest are those of Tanger-Tétouan-Al Hoceïma, Rabat-Salé-Kénitra and Béni Mellal- Khénifra, displaying altogether only 4.61 % of the total available potential.

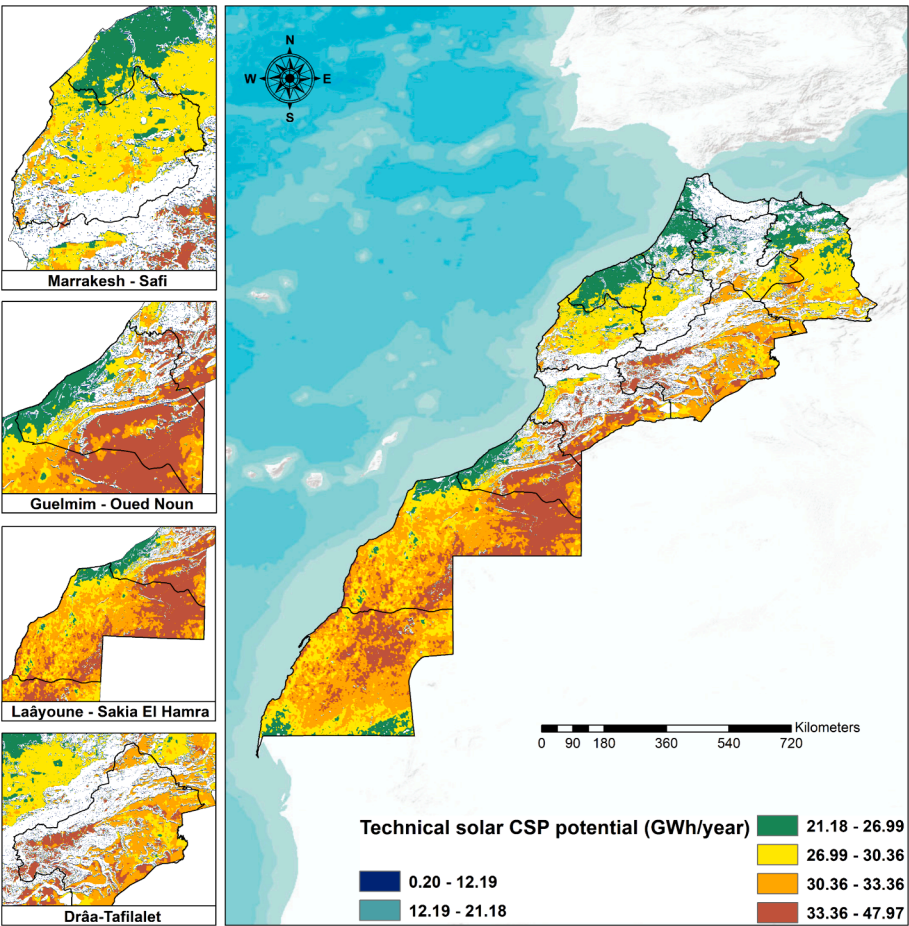


Fig. 7. Technical solar CSP potential overall map.

Table 2
Summary table of DNI values and technical solar CSP potential in regions of Morocco.

Region	Maximum DNI (kWh/m ² /year)	Average DNI (kWh/m ² /year)	Technical Potential (TWh/year)
Laäyoune-Sakia El Hamra	2626.00	2299.03	5549.72
Dakhla-Oued Ed-Dahab	2533.00	2203.44	5117.31
Drâa-Tafilalet	3284.00	2451.72	2250.49
L'Oriental	2832.00	2254.54	1923.92
Guelmim-Oued Noun	2922.00	2331.72	1594.48
Souss-Massa	3278.00	2472.61	1238.99
Marrakesh –Safi	3416.00	2281.16	1077.26
Fès-Meknès	2832.00	2249.35	707.00
Casablanca-Settat	2392.00	2154.04	669.29
Béni Mellal-Khénifra	3061.00	2291.08	469.11
Rabat-Salé-Kénitra	2617.00	2096.39	375.60
Tanger-Tétouan-Al Hoceïma	2729.00	2038.52	127.15
Sum			21 100.32

Comparative analysis of MCDM methods

Herein, site screening analysis started with weight calculations. The AHP weighting results obtained previously were combined with Buckley’s FAHP calculations, this time using two defuzzification methods (CoA and α -cut), with a view to identifying their impact on the final

Table 3
Exploitable surface and potential ratios per region.

Region	% of appropriate area to available area (%)	% of total exploitable area (%)	% of the total technical potential (%)
Laäyoune-Sakia El Hamra	98.21	25.96	26.30
Dakhla-Oued Ed-Dahab	98.69	25.55	24.25
Drâa-Tafilalet	58.31	9.65	10.67
L'Oriental	73.40	9.29	9.12
Guelmim-Oued Noun	78.16	7.28	7.56
Souss-Massa	50.61	5.25	5.87
Marrakesh –Safi	68.88	5.15	5.11
Fès-Meknès	46.00	3.40	3.35
Casablanca-Settat	89.46	3.46	3.17
Béni Mellal-Khénifra	42.71	2.24	2.22
Rabat-Salé-Kénitra	59.55	2.03	1.78
Tanger-Tétouan-Al Hoceïma	2.87	0.71	0.60
Sum		100 %	100 %

weight. Table 4 summarizes weighting results obtained by means of AHP and Fuzzy AHP (with CoA and α -cut) methods. As observed, a slight difference can be noticed between the overall weights and the detailed weights.

The overall criteria ranking follows the same logic for both selected

Table 4Weighting results using Fuzzy AHP method (using Geometric Mean Method) (with CoA and α -cut defuzzification method) and AHP method.

Criteria	Sub-Criteria	Using Fuzzy AHP (with CoA defuzzification)		Using Fuzzy AHP (with α -cut defuzzification)		Using AHP [61]	
		Detailed weights	Global weights	Detailed weights	Global weights	Detailed weights	Global weights
Climate	Direct Normal irradiation (DNI)	57.34 %	57.34 %	57.72 %	57.72 %	57.70 %	57.70 %
Orography	Slope	25.92 %	25.92 %	25.71 %	25.71 %	25.60 %	25.60 %
Water resources	Distance from water ways	9.30 %	11.67 %	9.21 %	11.54 %	9.36 %	11.70 %
	Distance from dams	2.38 %		2.33 %		2.34 %	
Location	Distance from cities and urban areas	1.42 %	5.07 %	1.40 %	5.04 %	1.40 %	5.00 %
	Distance from roads and railways	0.48 %		0.46 %		0.45 %	
	Distance from airports	0.60 %		0.59 %		0.60 %	
	Distance from Electricity grid	2.58 %		2.58 %		2.55 %	
	Sum	100 %	100 %	100 %	100 %	100 %	100 %

methods and is in line with the requirements of the CSP technology. Indeed, the most influential factor is climate, represented by the DNI, which reflects its relevance in a given site. In second place comes the orography represented by the slope which is also highly important from a techno-economic point of view, as stated before. The third place was given to the distance to the water resources, due to its importance for the operation and maintenance of the CSP, especially since the wet cooling configuration was chosen in this case study. Lastly, there is location in terms of infrastructure, since priority was given to the proximity to water resources to ensure reliable operation of the CSP plant.

Although CoA defuzzification method is the most widely used in literature [51], the weights- obtained by means of α -cut for climate and orography criteria were closer to AHP weights. However, the α -cut method generated a somewhat lower weighting for the criteria relating to water resources, as illustrated in Fig. 8.

As for consistency ratios (CR), all of them were satisfactory (below 0.1), which proves that pairwise comparisons were consistent and fulfilling. For AHP method the value amounted to 0.0285. Meanwhile, for Fuzzy AHP method, both the consistency of the mean values (CR^m) and the consistency of the lower and upper bounds (CR^g) were obtained and stood at 0.0324 and 0.0684 respectively.

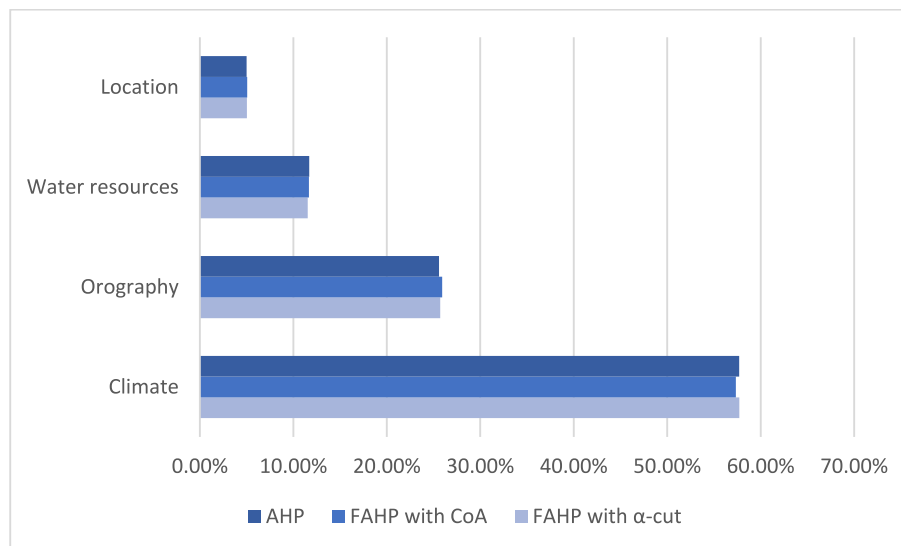
As a matter of fact, the criteria rankings remained unchanged although some differences that may seem unimportant can be clearly identified. However, in the upcoming section the impact of such value variations on the final ranking maps can be visualized and how the classes can vary in terms of coverage (area) can be demonstrated.

Comparison of the ranking maps

The aim of implementing the FAHP method is to reduce the subjectivity of the AHP method, and therefore the study went further than a mere comparison of weights to compile classification maps in order to spatially assess the spatial distribution of each class (5 classes from limited suitability to excellent) (see Fig. 9 and Fig. 10). Two defuzzification methods were used in order to gauge their impact, namely center of the area (CoA) and α -cut. ArcGIS tools allowed calculation of the ratio of each class in available surface area, in accordance with Fig. 11.

The discrepancy, no matter how slight, is noticeable when zooming in on the regions represented on the ranking maps, as depicted in Fig. 9 & Fig. 10. In fact, while comparing the three maps, the methods are in agreement on some excellent areas adequate for the CSP system especially in the center, the south eastern zones. The highly suitable zones were near the excellent ones more in the south and also in some zones in the east. AHP method has shown more surface for excellent zones especially in the south east and in the south. The suitable areas were more noticeable in the south western zones (near Tarfaya) and eastern areas, along with some surfaces in the west. Regarding the locations of limited suitability, they were mainly noticed in the northern east zones, some in the west a little away from shores, and between Sidi ifni and Akhefennir due to low DNI values around 1900 kWh/ m²/year.

In terms of area ratio, all the methods had roughly the same area ratio for the excellent class (AHP method yielded the maximum area (6 %), followed by the FAHP method with α -cut (Fig. 11)). FAHP method with CoA led to the highest area ratio for the highly suitable class and

**Fig. 8.** Weight comparison (%).

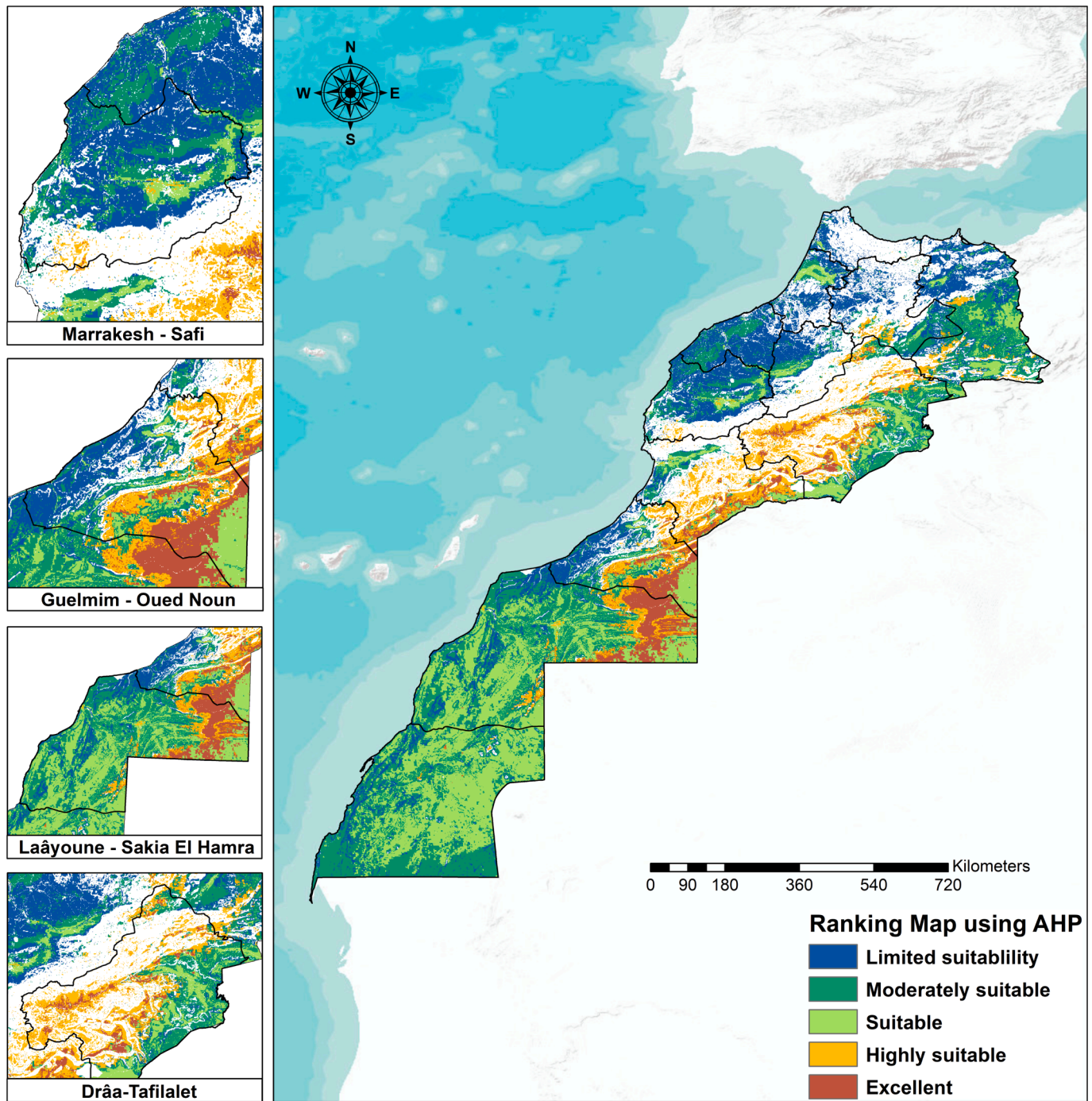


Fig. 9. Ranking map using AHP method.

the suitable class, with 9.77 % and 30.26 % respectively. As far as the “moderately suitable” class is concerned, the ratio was almost equal (varying between 29.6 % and 30.3 %). At last, the surface area ratio rankings were similar to the excellent class (the AHP method resulting in a maximum of 20.08 %).

To summarize, FAHP with CoA defuzzification method have yielded more area ratio in highly suitable and suitable classes and the minimum of all for the limited suitability class. Meanwhile The FAHP with α -cut

defuzzification and AHP methods were more closely matched in both values and variation across all classes. One can imply that α -cut defuzzification is closer to subjectivity than the CoA method, which showed more discrepancies. With this classification, AHP favored slightly more excellent areas than the other FAHP method outcomes. Alongside certain areas scored slightly lower to be considered highly appropriate (Fig. 11). It can be asserted that the FAHP method with CoA can help reduce subjectivity more than α -cut defuzzification.

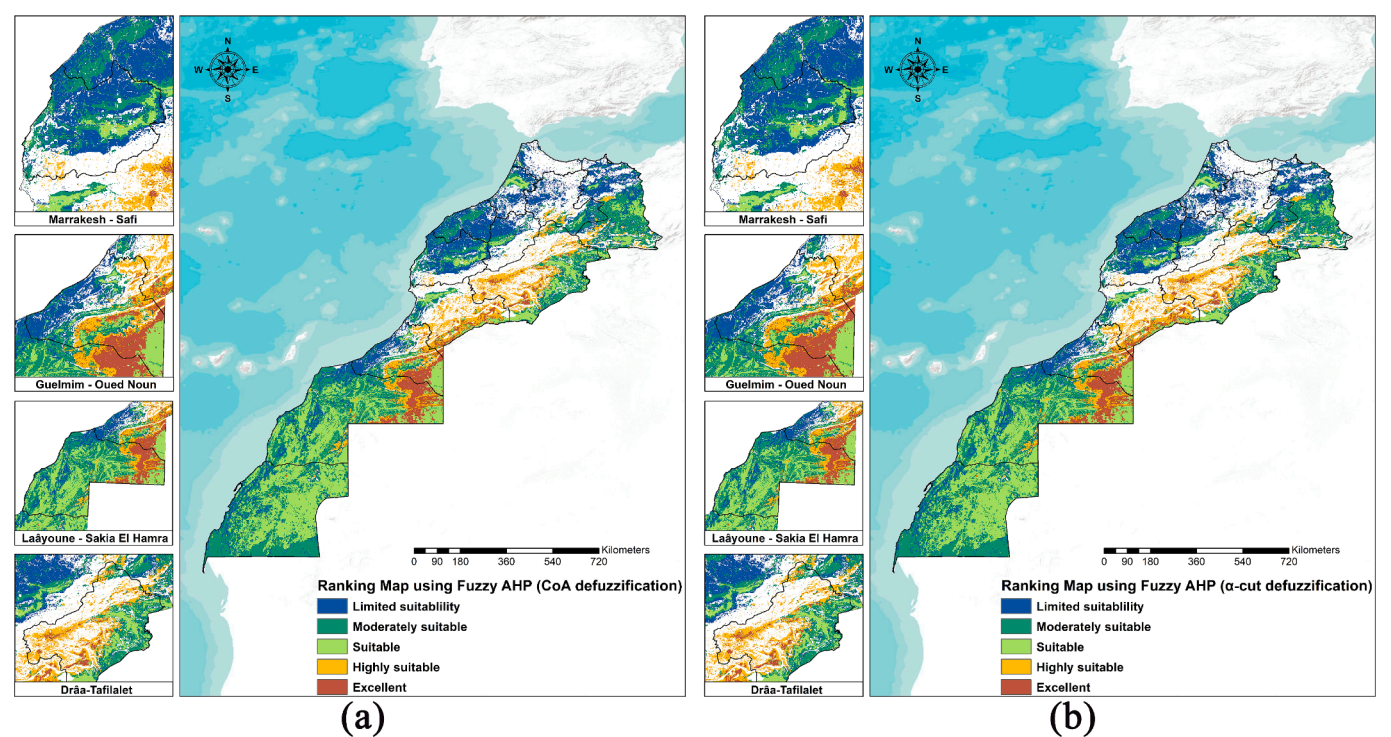


Fig. 10. Ranking maps: (a) using FAHP (with CoA defuzzification method); (b) using FAHP (with α -cut defuzzification method).

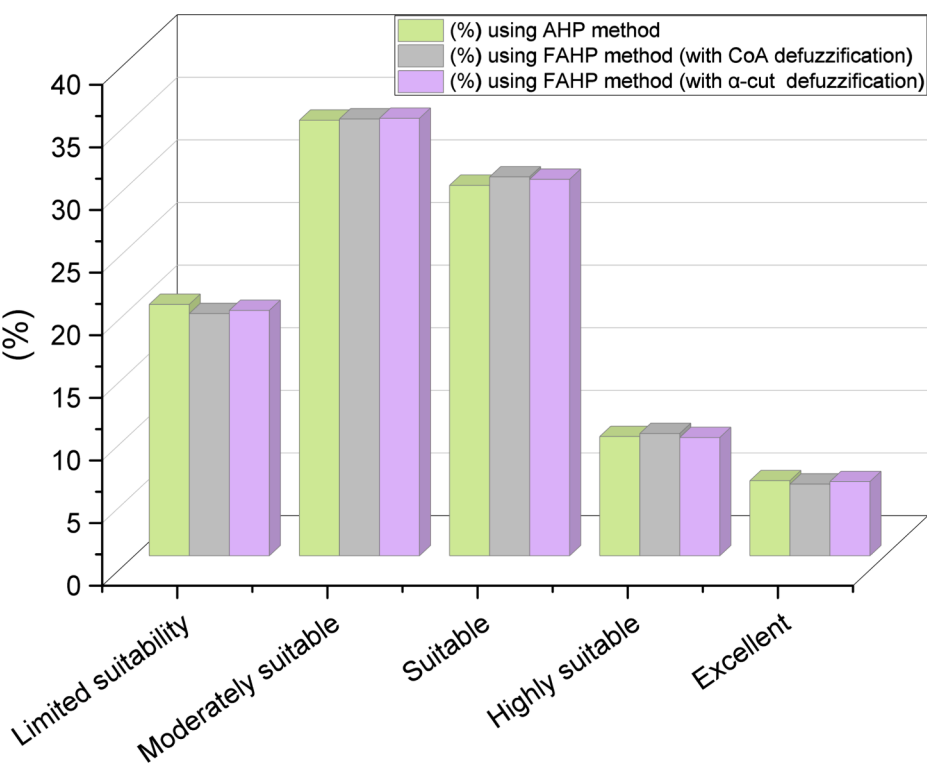


Fig. 11. Comparison of surfaces (%).

Conclusion

Throughout this work, an assessment of the CSP solar technical potential existing in Morocco together with a comparative site selection analysis were performed with a view to flagging and categorizing adequate sites, thereby gauging the impact of introducing fuzziness to the AHP method, with regard to defuzzification methods simultaneously.

Potential appraisal and evaluation relied on the approach of Hermann et al, with a formula depending on the solar resource represented by the DNI, the plant efficiency varying along the DNI, the spacing factor and the available surface area. The latter was assessed by eliminating unfit areas altogether, mainly with regard to both land occupation and technical limitations relative to CSP technology. The technical potential map was derived from the potential per cell rising from 0.20 to 47.97 GWh/year. The overall CSP technical potential stood at 21,100.32 TWh/year representing a highly promising amount. Regionally, most areas tend to be extremely propitious, notably the southern regions, with a few in the center and the east. By contrast, the northern regions show somewhat less favorable levels for CSP installations, chiefly owing to lower DNI values. The region of Laâyoune-Sakia El Hamra and Dakhla-Oued Ed-Dahab lead the way in terms of potential, reaching 5549.72 and 5117.31 TWh/year respectively. Alongside this, regions with the lowest potential were Rabat-Salé-Kénitra and Tanger-Tétouan-Al Hoceïma, totaling only 375.60 and 127.15 TWh/year accordingly.

Thereafter, a comparative site screening analysis of the two Multi Criteria Decision Making (MCDM) methods –Analytic Hierarchy Process (AHP) and Fuzzy AHP – was performed with a view to discerning discrepancies regarding weighting maps and rankings. Particular emphasis is placed on spatial distribution (% area ratio) providing insight into the weight and spatial variation within each class. The investigation follows the path of usual site selection studies starting from case study focus, and exclusion mask generation, to weight calculus, to finally create three ranking maps depicting suitable zones based on AHP and FAHP methods outcomes. In line with existing literature, screening criteria were structured according to climate, orography, proximity to water resources and location. Weight computation revealed roughly the same weighting with minor variations, whereby the α -cut was more closely aligned with the AHP weights (Table 4). Indeed, the overall ranking was alike for both methods given that the same logic was applied upon classification: the climate represented by DNI was ranked first (57.72 % for FAHP with α -cut, 57.34 % for FAHP with CoA, 57.70 % for AHP), followed by orography – represented by slope criterion- as CSP is slope-sensitive (25.71 % for FAHP with α -cut, 25.92 % for FAHP with CoA, 25.6 % for AHP). Proximity to water resources (rivers and dams) was ranked third due to the importance of water availability, especially regarding the wet cooling configuration at issue (11.54 % for FAHP with α -cut, 11.67 % for FAHP with CoA, 11.70 % for AHP). Location criteria were ranked fourth with the following weights: (5.04 % for FAHP with α -cut, 5.07 % for FAHP with CoA, 5.00 % for AHP).

The constraining zones were determined afterwards along with the buffers for merge. The areas are either non-harvestable due to occupation/environmental restrictions or unsuitable from a techno-economic viewpoint. Areas under scrutiny were zones with a steep slope, protected areas and forests, residential areas, infrastructure (roads, railways, grid network), water bodies and military zones. Once collected and merged using ArcGIS software, the mask was processed and subtracted before generating the final ranking maps. The latter ones took into account weightings resulting from each method. The final maps

depict adequate zones to host wet cooling CSP power plants sorted in ten classes (5 classes ranging from limited suitability to excellent). This rating gives an overview of the sites, allowing an assessment of potential deployment sites for solar thermal power plants. FAHP with CoA defuzzification gave a higher area ratio in both the highly suitable and suitable classes, and the minimum of all for the limited suitability class. On the other hand, the FAHP methods with α -cut defuzzification and AHP were closest to each other in terms of values and variations across all classes. This suggests that α -cut defuzzification provides a close approximation to subjectivity than the CoA method, which showed more variation.

Ultimately, this investigation has allowed a thorough potential assessment with method comparison of two methods AHP and Buckley's FAHP (with two defuzzification methods) in an energy context, providing valuable data, especially the visualization of the suitability maps. The effect of fuzziness is shown at the level of ranking maps and we could discern the defuzzification method that allowed more reduction the subjectivity (FAHP with CoA defuzzification).

The present investigation's findings may assist researchers and policy makers to enforce the used method, while ensuring that both can be used with certain purposes, one of them might be heat generation for the industrial sector. Further feasibility studies, as well as the calculation of the Levelized cost of electricity (LCOE), can be carried out to enable the practical implementation of solar thermal power plant projects. Alongside all this, the authors might apply the FAHP method in the future for different RE systems and applications site selection studies.

Author statement

During the preparation of this work the author(s) declare that they did not use any Generative AI or AI-assisted technologies in the writing process.

CRedit authorship contribution statement

Ouafae Jbahi: Writing – original draft, Visualization, Software, Formal analysis, Data curation, Conceptualization. **Fatima-Zahra Ouchani:** Writing – original draft, Visualization, Software, Data curation. **Abdellatif Ghennioui:** Writing – review & editing, Supervision. **Mohammed Ferfra:** Writing – review & editing, Supervision. **Mohamed Cherkaoui:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendices

Table A1
Linguistic Variables for Fuzzy Pairwise Scale [36]

Linguistic Scale	Triangular Fuzzy Scale	Triangular Fuzzy Reciprocal Scale
Equally important	$\tilde{1} = (1,1,1)$	$\tilde{1}^{-1} = (1,1,1)$
Judgment values between equally and moderately	$\tilde{2} = (1,2,3)$	$\tilde{2}^{-1} = (1/3,1/2,1)$
Moderately more important	$\tilde{3} = (2,3,4)$	$\tilde{3}^{-1} = (1/4,1/3,1/2)$
Judgment values between moderately and strongly	$\tilde{4} = (3,4,5)$	$\tilde{4}^{-1} = (1/5,1/4,1/3)$
Strongly more important	$\tilde{5} = (4,5,6)$	$\tilde{5}^{-1} = (1/6,1/5,1/4)$
Judgment values between strongly and very strongly	$\tilde{6} = (5,6,7)$	$\tilde{6}^{-1} = (1/7,1/6,1/5)$
Very strongly more important	$\tilde{7} = (6,7,8)$	$\tilde{7}^{-1} = (1/8,1/7,1/6)$
Judgment values between very strongly and extremely	$\tilde{8} = (7,8,9)$	$\tilde{8}^{-1} = (1/9,1/8,1/7)$
Extremely more important	$\tilde{9} = (8,9,9)$	$\tilde{9}^{-1} = (1/9,1/9,1/8)$

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